

Engineers Without Borders - SVCE Student Chapter

Initiative of the year 2025-26

Carbon Footprints

"The Earth doesn't need us to save it, it needs us to stop harming it."

-Start with a Switch-



TABLE OF CONTENTS

S.No	TITLE	PAGE NO
1.	EXECUTIVE SUMMARY	3
2.	OBJECTIVES	4
3.	IMPACT ASSESSMENT	4
4.	ALIGNMENT WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs)	4
5.	GALLERY	5
6.	CONCLUSION	9
7.	FUTURE SCOPE	9

Executive Summary

The carbon footprint awareness initiative by the SVCE Student Chapter of Engineers Without Borders (EWB) is aimed at encouraging students to adopt responsible electricity usage and develop a deeper awareness of their environmental impact. The study focuses on commonly used classroom appliances such as lights, fans, and projectors, analyzing how their daily usage contributes to carbon emissions.

The assessment is based on key factors such as appliance power ratings, duration of usage, and typical classroom usage patterns. By comparing two practical scenarios, one where appliances are left ON regardless of necessity, and another where they are switched OFF during breaks or after lectures the study clearly demonstrates the impact of user behavior on energy consumption and carbon emissions.

The results show that unnecessary usage of electrical devices leads to a significant increase in carbon emissions, while simple actions like switching off appliances when not in use can considerably reduce energy waste. For example, lights and fans, which are used continuously throughout the day, contribute a noticeable share of emissions when left running unattended. However, adopting mindful habits can bring down these emissions without affecting comfort or productivity.

An important takeaway from this initiative is that sustainability does not always require complex solutions or advanced technology. Instead, consistent small actions such as turning off switches, being aware of energy usage, and encouraging others to do the same can collectively create a substantial positive impact when practiced across classrooms and institutions.

This initiative strongly aligns with the principles of sustainable development by promoting energy efficiency, reducing greenhouse gas emissions, and fostering a culture of environmental responsibility among students. It reinforces the message that “every action counts” and highlights the role of individuals in contributing to larger climate goals.

Overall, this effort serves as both an awareness model and a call to action, motivating students and educational institutions to integrate simple, energy-efficient practices into their daily routines. By doing so, it contributes to building a more sustainable, responsible, and environmentally conscious future.

OBJECTIVES:

The primary objectives of the Carbon Footprint Awareness Initiative were:

- To analyze the carbon emissions generated by commonly used classroom appliances such as lights, fans, and projectors.
- To evaluate the impact of appliance usage patterns, including duration and frequency, on overall energy consumption.
- To compare carbon emissions under two conditions: when appliances are used continuously (“ON”) and when they are switched off during non-essential periods.
- To create awareness among students about the importance of energy conservation and its direct link to reducing carbon footprint.
- To promote simple and practical behavioral changes, such as switching off appliances when not in use, to minimize unnecessary energy wastage.
- To encourage students to adopt responsible and sustainable energy usage habits in their daily routines.
- To demonstrate how small individual actions can collectively contribute to significant environmental benefits.
- To align student activities with sustainable development goals by reducing greenhouse gas emissions and promoting environmental responsibility.

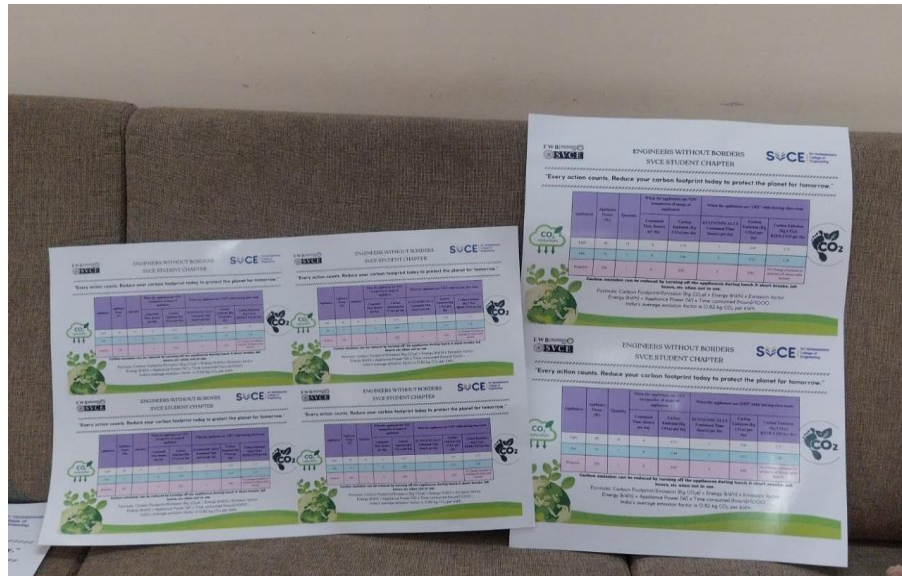
IMPACT ASSESSMENT:

- Increased awareness about carbon emissions from everyday appliance usage.
- Promoted responsible energy usage and reduced unnecessary power consumption.
- Encouraged simple habits like switching off appliances during non-use.
- Fostered environmental responsibility and sustainable behavior among students.
- Contributed to lowering the overall carbon footprint of the institution.

ALIGNMENT WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs):



VISUAL DOCUMENTATION:



ENGINEERS WITHOUT BORDERS
SVCE STUDENT CHAPTER

"Every action counts. Reduce your carbon footprint today to protect the planet for tomorrow."

Appliances	Appliance Power (W)	Quantity	When the appliances are 'ON' irrespective of usage of appliances		When the appliances are 'OFF' while leaving class room		
			Consumed Time (hours) per day	Carbon Emission (Kg CO ₂ e) per day	Economically Consumed Time (hours) per day	Carbon Emission (Kg CO ₂ e) per day	Carbon Emission (Kg CO ₂ e) REDUCED per day
Light	40	18	8	4.72	5	2.95	1.77
Fan	75	7	8	3.44	5	2.15	1.29
Projector	250	1	4	0.82	4	0.82	No Change (Assumed as switched off always after lecture)

Carbon emission can be reduced by turning off the appliances during lunch & short breaks, lab hours, etc when not in use.
Formula: Carbon Footprint/Emission (Kg CO₂e) = Energy (kWh) x Emission factor
Emission factor = Power (W) x Time consumed (hours)/1000



Block 2 - Snaps

EWB SVCE ENGINEERS WITHOUT BORDERS
SVCE STUDENT CHAPTER

"Every action counts. Reduce your carbon footprint today to protect the planet for tomorrow."

Appliances	Appliance Power (W)	Quantity	When the appliances are 'ON' irrespective of usage of appliances		When the appliances are 'OFF' 40% less than class room		
			Consumed Time (hours) per day	Carbon Emission (kg CO ₂ e) per day	Economically Consumed Time (hours) per day	Carbon Emission (kg CO ₂ e) per day	Carbon Emission (kg CO ₂ e) REDUCED per day
Light	40	15	8	3.92	5	2.46	1.48
Fan	75	7	8	3.44	5	2.15	1.29
Projector	250	1	4	0.82	4	0.82	No Change (Assumed as switched off always after lecture)

Carbon emission can be reduced by turning off the appliances during lunch & short breaks, lab hours, etc when not in use.
Formula: Carbon Footprint/Emission (kg CO₂e) = Energy (kWh) x Emission factor
Energy (kWh) = Appliance Power (W) x Time consumed (hours)/1000 ;
India's average emission factor is 0.82 kg CO₂ per kWh

Block 3 - Snaps





ENGINEERS WITHOUT BORDERS
SVCE STUDENT CHAPTER



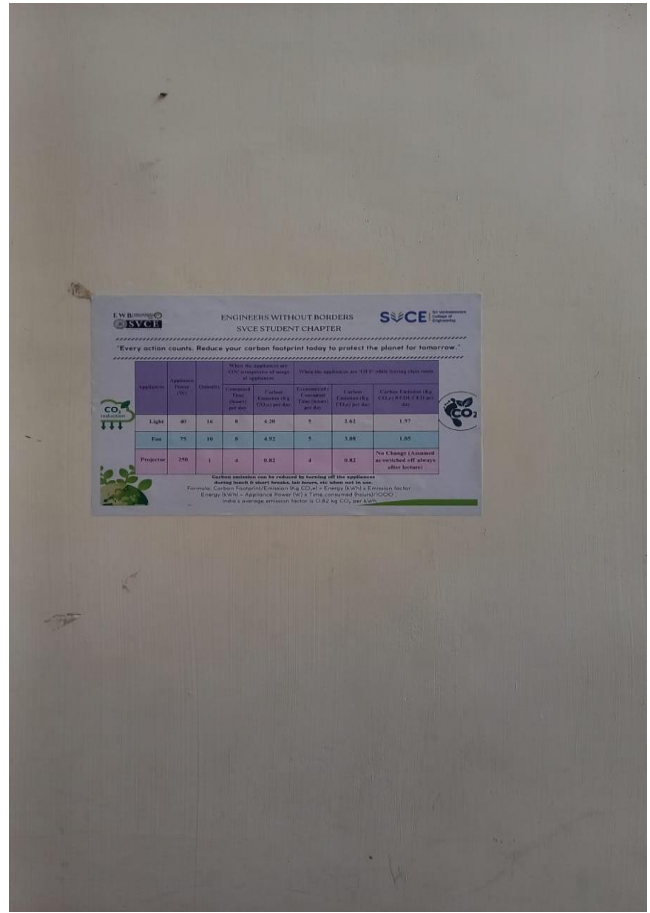
SVCE
SVKM'S Institute of
Technology

"Every action counts. Reduce your carbon footprint today to protect the planet for tomorrow."

Appliances	Appliance Power (W)	Quantity	When the appliances are 'ON' irrespective of usage of appliances		When the appliances are 'OFF' while leaving class room		
			Consumed Time (hours) per day	Carbon Emission (kg CO ₂ e) per day	Economical Consumed Time (hours) per day	Carbon Emission (kg CO ₂ e) per day	Carbon Emission, kg CO ₂ e REDUCED per day
Light	40	18	8	4.72	5	2.95	1.77
Fan	75	7	8	3.44	5	2.15	1.29
Projector	250	1	4	0.82	4	0.82	No Change (Assumed as switched off always after lecture)

Carbon emission can be reduced by turning off the appliances during lunch & short breaks, lab hours, etc when not in use.
 Formula: Carbon Footprint/Emission (kg CO₂e) = Energy (kWh) × Emission factor
 Energy (kWh) = Appliance Power (W) × Time consumed (hours)/1000;
 India's average emission factor is 0.82 kg CO₂ per kWh

Block 4 - Snaps



Block 5 – Snaps



CONCLUSION

This carbon footprint awareness initiative shows that reducing environmental impact does not always require complex solutions. Instead, it begins with simple, everyday actions like switching off lights, fans, and projectors when they are not needed. The study clearly highlights how small behavioral changes can significantly reduce energy consumption and carbon emissions in classroom environments.

Through this initiative, students become more aware of how their daily habits affect the environment. It encourages a sense of responsibility and reminds us that each individual action, no matter how small, contributes to a larger impact when practiced collectively.

Overall, this effort demonstrates that building a sustainable future starts with mindful choices in our daily lives. By adopting these simple practices consistently, students and institutions can work together to create a greener, more energy-efficient, and environmentally responsible campus.

FUTURE SCOPE

In the future, this initiative can be expanded by introducing smart energy-saving systems in classrooms and promoting paperless practices. Further awareness programs and small-scale renewable energy solutions can also be explored to reduce carbon emissions. These steps can help in building a more sustainable and eco-friendly campus environment